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Pd(0)-Catalyzed Coupling–Cyclization of 2-(2′,3′-Allenyl)acetylacetates and Organic Halides: An Efficient Synthesis of 4,5-Dihydrofuran Derivatives

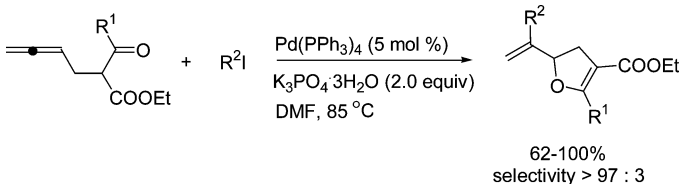
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ABSTRACT



Highly chemo- and regioselective Pd(0)-catalyzed coupling–cyclization of 2-(2′,3′-allenyl)acetylacetates with organic halides using K_3PO_4 as base in DMF efficiently afforded 4,5-dihydrofuran derivatives in moderate to excellent yields, with a selectivity of >97:3.

The development of new chemical processes for producing elaborate and important hetereocyclic structures in an efficient manner has become an important area of research in organic chemistry due to the importance of heterocyclic compounds.[1] Among heterocyclic compounds, dihydrofuran derivatives are intensively studied. They are important subunits in many biologically active compounds[2] and are also important intermediates in organic synthesis.[3] Thus, much attention has been paid to the development of new methods for the synthesis of furans and dihydrofurans.[4]

Recently, we and others have shown that functionalized allenes can be facile starting materials for the preparation of carbocycles and heterocycles.[5,6] Especially, 2-(2′,3′-allenyl)malonates could afford carbocycles, i.e., vinylic cyclopropane[7a] and cyclopentene[7b] derivatives, highly selectively by tuning the solvent and base used; the reaction of 1,2-allenyl ketones and organic halides in toluene using Et_3N as the base affords polysubstituted furans by the catalysis of Pd(0) and Ag_2CO_3;[7c−e] 2,5-dihydrofurans and 5,6-dihydropyrans can be synthesized via the Pd(II)-catalyzed coupling−cyclization of allylic halides with 2,3- or 3,4-allenols, respectively.[7f,g] The reaction of 3,4-allenols with aryl iodides afforded 2,3-dihydrofurans via the oxidative addition−*exo*-mode oxypalladation−reductive elimination sequence.[7h] Based on these observations, we envisioned that a Pd(0)-catalyzed cyclization of an organic halide with

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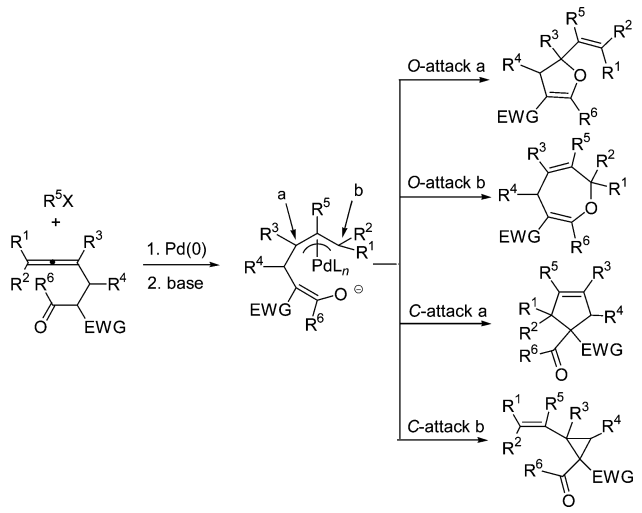
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2-(2′,3′-allenyl)acetylacetates may form 4,5-dihydrofurans, 2,5-dihydroxepines, cyclopentenes, or cyclopropanes with the intramolecular attack of the enolate intermediate (Scheme 1). In this paper, we report a highly selective formation of

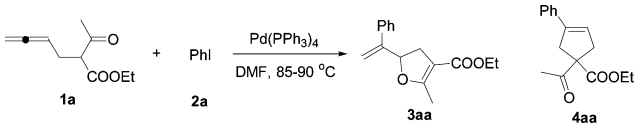
Scheme 1



4,5-dihydrofurans from organic halides and 2-(2′,3′-allenyl)-acetylacetates.

At first, a mixture of 2-(2′,3′-allenyl)acetylacetate **1a**, PhI, and LiOH in DMF was stirred in the presence of $Pd(PPh_3)_4$ at 85 °C for 7 h to afford the expected product **3aa** in 63% yield together with the formation of the five-membered carbocycle **4aa** in 15% yield (entry 1, Table 1). To improve the selectivity and yield, the base effect was studied (Table 1). With the addition of LiOH, *t*-BuOK, or NaOH, the reaction afforded a mixture of **3aa** and **4aa** (entries 1−3, Table 1). Using bases such as NaH (entry 4, Table 1), K_2CO_3 (entry 5, Table 1), Et_3N (entry 6, Table 1), KOH (entry 7, Table 1), or K_3PO_4 (entry 9, Table 1), the reaction afforded **3aa** as the only product. Without base, the reaction is also highly selective, but product **3aa** was formed in only 13% isolated yield (entry 8, Table 1). K_3PO_4 turned out to be the best base, affording **3aa** in a quantitive yield.

Table 1. Effect of Base on the Pd(0)-Catalyzed Coupling−Cyclization of 2-(2′,3′-Allenyl)acetylacetate **1a** and PhI[a]



entry	base	time (h)	isolated yield of **3aa** (**4aa**) (%)	ratio **3aa**:**4aa**[b]
1	LiOH	7	63 (15)	81:19
2	*t*-BuOK	8	54	91:9
3	NaOH	8	68	99:1
4	NaH	7	45	100:0
5	K_2CO_3	8	68	100:0
6	Et_3N	7	73	100:0
7	KOH	8	68	100:0
8		8	13[c]	100:0
9	**$K_3PO_4 \cdot 3H_2O$**	**8**	**100**	**100:0**

[a] The reaction was conducted using **3aa** (0.15 mmol), $Pd(PPh_3)_4$ (5 mol %), PhI (1.2 equiv), and base (2.0 equiv) in DMF. [b] Determined by NMR analysis. [c] 71% of **1a** was recovered.

The solvent effect using K_3PO_4 as base was then studied. THF (entry 2, Table 2) and CH_3CN (entry 3, Table 2) formed product **3aa** together with 2−3% of **4aa.** Toluene (entry 4, Table 2), DMSO (entry 5, Table 2), 1,4-dioxane (entry 6, Table 2), or CH_2Cl_2 (entry 7, Table 2) afforded product **3aa** as the only product in relative low yields. The reaction in CH_3OH (entry 8, Table 2) failed to afford the products. DMF is the best solvent (entry 1, Table 2).

By applying these standard reaction conditions, we studied the synthesis of 2-methyl-3-ethoxycarbonyl-5-vinyl-4,5-dihydrofuran from 2-(2′,3′-allenyl)acetylacetate **1a** and different organic halides. The results are summarized in Table 3. From the results in the table, the following issues should

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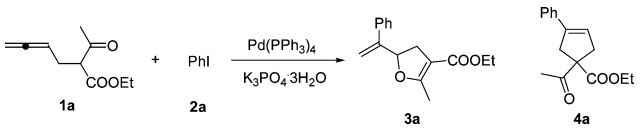
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Table 2. Solvent Effect of Pd(0)-Catalyzed Coupling−Cyclization of 2-(2′,3′-Allenyl)acetylacetate **1a** and PhI[a]

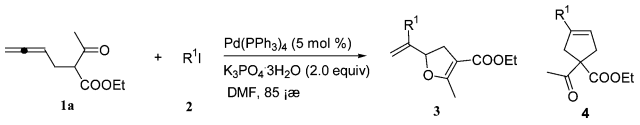


entry	solvent	temp (°C)	time (h)	isolated yield of **3aa** (%)	ratio **3aa**:**4aa**
1	**DMF**	**85**	**8**	**100**	**100:0**
2	THF	reflux	8	81	100:0
3	CH_3CN	reflux	8	81	≥97:3
4	toluene	85	8	91	100:0
5	DMSO	85	8	82	100:0
6	dioxane	85	8	87	100:0
7	CH_2Cl_2	reflux	6	78	100:0
8	CH_3OH	reflux	6	0	

[a] The reaction was conducted using **1a** (0.15 mmol), $Pd(PPh_3)_4$ (5 mol %), PhI (1.2 equiv), and K_3PO_4 (2.0 equiv).

be noted: (i) the yields of this reaction range from moderate to excellent; (ii) electron-rich (entries 2 and 3, Table 3),

Table 3. Reaction of **1a** with Different Organic Iodides

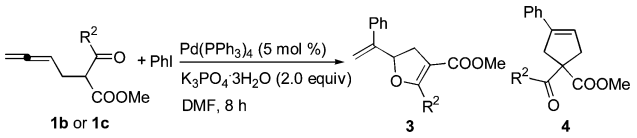


entry	R^1	product	time (h)	isolated yield of **3a** (%)	ratio **3a**:**4a**
1	Ph	**3aa**	8	100	100:0
2	4-MeC_6H_4	**3ab**	8	87	100:0
3	4-$MeOC_6H_4$	**3ac**	8	89	100:0
4	4-$MeO_2CC_6H_4$	**3ad**	8	58	99:1
5	4-AcC_6H_4	**3ae**	2	100	97:3
6	4-BrC_6H_4	**3af**	8	80	99:1
7	1-naphthyl	**3ag**	8	77	100:0
8	1-(*E*)-hexenyl	**3ah**	7	81	100:0
9	2-thienyl	**3ai**	2	84	99:1

electron-deficient (entries 4 and 5, Table 3), Br-substituted (entry 6, Table 3) phenyl halides, and naphthyl (entry 7, Table 3), hexenyl (entry 8, Table 3), and thienyl (entry 9, Table 3) iodides all afforded the corresponding 4,5-dihydrofurans with high selectivity.

Different 2-(2′,3′-allenyl)acetylacetates can also be used in this reaction (Table 4).

Table 4. Steric Effect in the Reaction of 2-(2',3'-Allenyl)acetylacetates with Phenyl Iodide

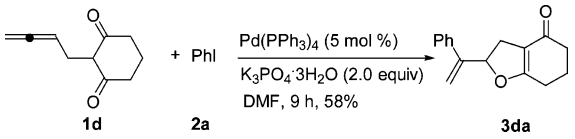


1	temp (°C)	product	isolated yield (%)		ratio **3**:**4**[a]
1b (R = *n*-Pr)	85	**3ba**	91	**4ba**	99:1
1c (R = *i*-Pr)	85	**3ca**	81	**4ca**	98:2
1c	60	**3ca**	40	**4ca**	97:3
1c	100	**3ca**	40	**4ca**	97:3

[a] Determined by NMR analysis.

The reaction of 2-(2′,3′-butadienyl)cyclohexane-1,3-dione afforded the bicyclic product **3da**, providing an opportunity for the construction of a bicyclic skeleton (Scheme 2).

Scheme 2. Reaction of 2-(Buta-2′,3′-dienyl)cyclohex-1,3-dione and Phenyl Iodide



In conclusion, we have developed an efficient method for the synthesis of substituted 4,5-dihydrofurans with different substitution patterns in high chemo- and regioselectivity. The study of new chemo-, regio-, and stereoselective reactions for differently substituted 2-(2′,3′-allenyl)acetylacetates are currently being carried out in our laboratory.

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Supporting Information Available: Analytical data for all products not listed in the text. This material is available free of charge via the Internet at http://pubs.acs.org.

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